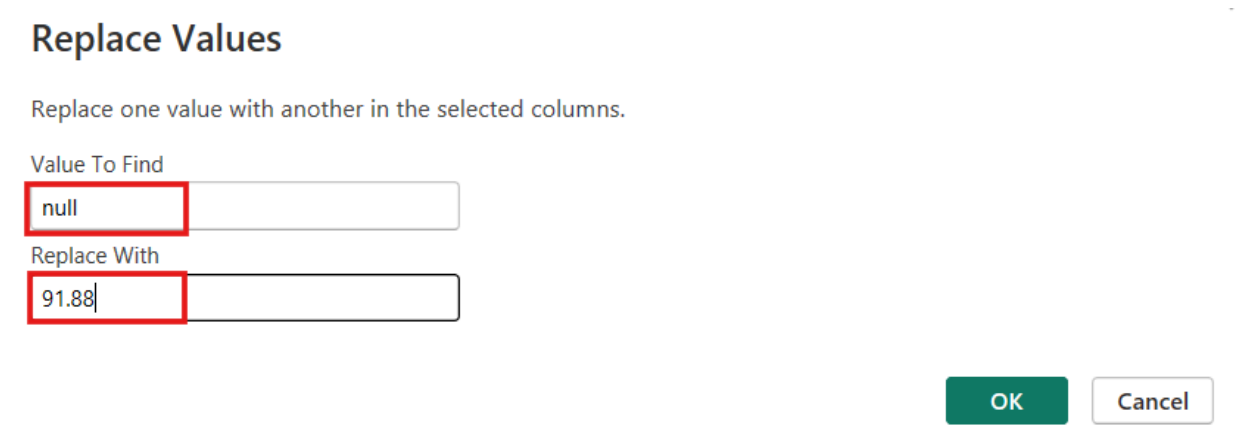
Power BI Module End Assignment

1. Data Cleaning & Modeling:

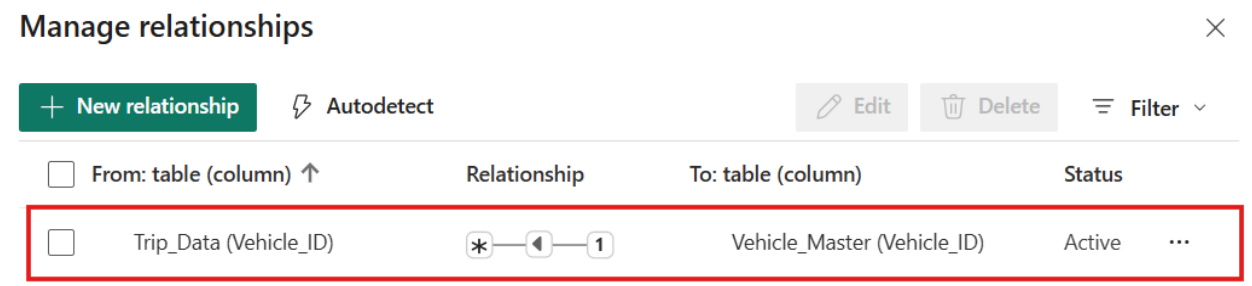
○ Fix missing fuel consumption values (use mean imputation).

Fuel Consumption contains **6% null values**. The mean value was calculated using the **Statistics → Average** option, resulting in a mean of **91.88**. The null values were then replaced with the calculated mean value using **Home → Replace Values** in Power Query.



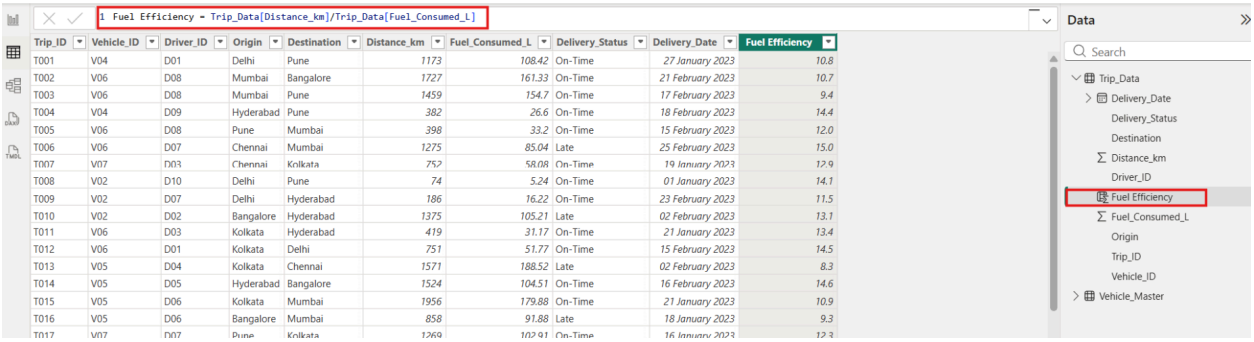


○Created a relationship between the **Trip_Data** and **Vehicle Master** tables using the **Vehicle_ID** field.



2. DAX Measures:

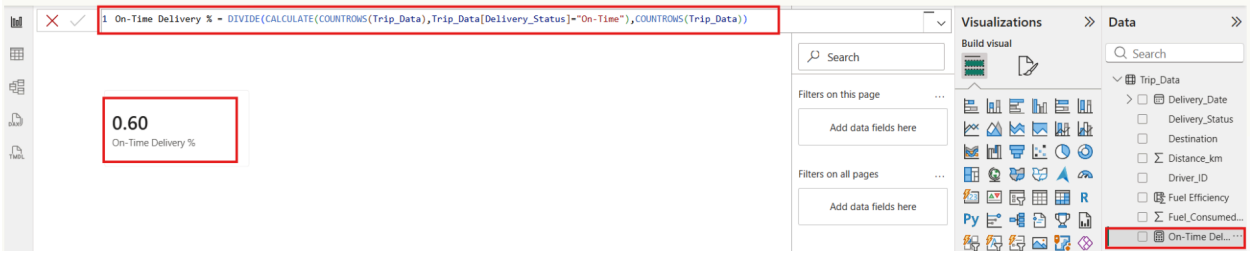
○ Fuel Efficiency = Distance / Fuel Consumed



Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date	Fuel Efficiency
T001	V04	D01	Delhi	Pune	1173	108.42	On-Time	27 January 2023	10.8
T002	V06	D08	Mumbai	Bangalore	1727	161.33	On-Time	21 February 2023	10.7
T003	V06	D08	Mumbai	Pune	1459	154.7	On-Time	17 February 2023	9.4
T004	V04	D09	Hyderabad	Pune	382	26.6	On-Time	18 February 2023	14.4
T005	V06	D08	Pune	Mumbai	398	33.2	On-Time	15 February 2023	12.0
T006	V06	D07	Chennai	Mumbai	1275	85.04	Late	25 February 2023	15.0
T007	V07	D03	Chennai	Kolkata	752	58.08	On-Time	19 January 2023	12.9
T008	V02	D10	Delhi	Pune	74	5.24	On-Time	01 January 2023	14.1
T009	V02	D07	Delhi	Hyderabad	186	16.22	On-Time	23 February 2023	11.5
T010	V02	D02	Bangalore	Hyderabad	1375	105.21	Late	02 February 2023	13.1
T011	V06	D03	Kolkata	Hyderabad	419	31.17	On-Time	21 January 2023	13.4
T012	V06	D01	Kolkata	Delhi	751	51.77	On-Time	15 February 2023	14.5
T013	V05	D04	Kolkata	Chennai	1571	188.52	Late	02 February 2023	8.3
T014	V05	D05	Hyderabad	Bangalore	1524	104.51	On-Time	16 February 2023	14.6
T015	V05	D06	Kolkata	Mumbai	1956	179.88	On-Time	21 January 2023	10.9
T016	V05	D06	Bangalore	Mumbai	858	91.88	Late	18 January 2023	9.3
T017	V07	D07	Pune	Kolkata	1269	102.91	On-Time	16 January 2023	12.3

○On-Time Delivery % was calculated by dividing the **number of On-Time Trips by the Total Number of Trips**, using the following DAX measure.

On-Time Delivery % =
DIVIDE(CALCULATE(COUNTROWS(Trip_Data),Trip_Data[Delivery_Status]="On-Time"),COUNTROWS(Trip_Data))



○ Cost per km was calculated by dividing the **total Fuel Cost and Maintenance Cost by the Distance travelled**, using the following DAX measure.

```
Cost per km =
DIVIDE(100+SUM(Vehicle_Master[Maintenance_Cost]),SUM(Trip_Data[Distance_km]))
```



3. Visualization:

Created a **Bar Chart** to display **On-Time Delivery % by Destination**.
Created a **Line Chart** to visualize the **Fuel Efficiency Trend by Delivery Date**.
Created a **Card Visualization** to display the **Average Cost per Km**.
Created a **Pie Chart** to represent the **Average Maintenance Cost by Vehicle Type**.



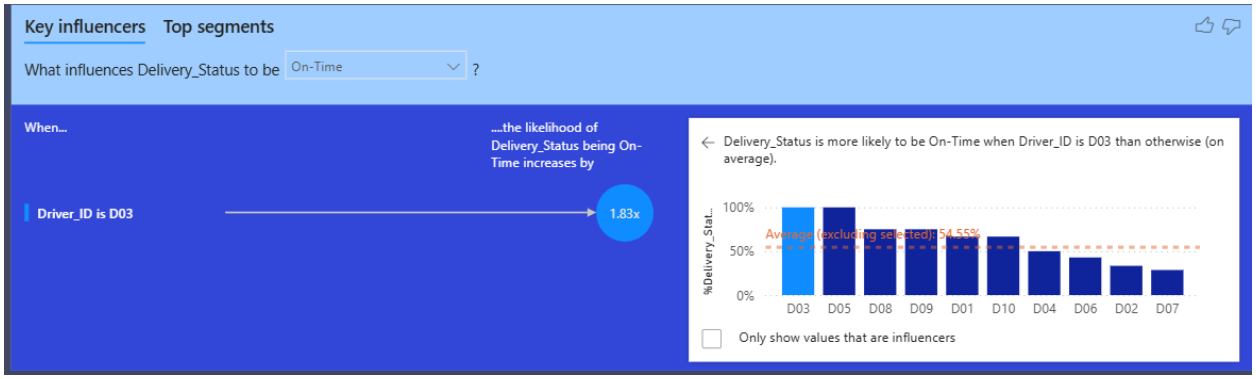
4. AI-Powered Visuals

❖ Q&A Visual:
➢ Add Q&A visual → Prompt: “Average Cost per km by vehicle type?”

Unable to create the **Q&A Visual**, as this feature is currently unavailable in Power BI.

❖ Key Influencers Visual:
➢ Add Delivery Status in the Analyze and explain by -distance in km,vehicle_type, Driver_ID

The **Key Influencers visual** was created based on the given requirement. The analysis indicates that **Driver ID D03 is an influencing factor for On-Time Delivery**.



❖ Decomposition Tree (AI Visual):
➢ Analyze Cost per km → explained by Vehicle Type, Maintenance_cost, and Distance_km.

The **Decomposition Tree (AI Visual)** was created based on the given requirement to analyze and break down the data across different dimensions.



Expected Output: A transport operations dashboard to optimize routes and fleet usage.